Brown University Computer Science PhD Program: Personal Statement

As legal tools are lagging in mitigating the harmful impacts of increasingly capable systems, computational tools are being wielded in the name of legal values. Throughout my undergraduate studies in computer science and mathematics, I witnessed how this feedback loop created opportunities for subversion. Consequently, I was motivated to develop research skills at the intersection of computation and law within MIT's technology and policy master's program. As a PhD student in computer science at Brown University, I would be eager to examine how computation and law could, together, drive the design and development of technical systems toward equitable and intended ends. I would be excited to 1) leverage domain knowledge to co-design models that support legal processes and frameworks; 2) uncover failures in computational systems and in their development; and 3) propose and motivate system standards and technology regulation.

Research Motivation. During an engineering internship at Google in 2020, I worked on open-source differential privacy (DP) libraries under Chris Jones, Raine Serrano, and Miraç Basaran. Differential privacy strategically noises data to achieve privacy guarantees while retaining information. For data in histogram publication, however, adding noise is not enough. Consider a health survey where a rare and sensitive response is linked to a single participant. The appearance of that response as a histogram partition, regardless of noise to the partition's contribution, would leak the presence of the participant in the database. This violates DP guarantees. Thus, δ-thresholding was used throughout Google's codebase to prevent partition leakage. However, with thresholding, partitions with small contributions that are predetermined and public were also needlessly dropped[1]. I designed and modified the aggregation functions in the codebase so that developers could use the library to specify and keep public, predetermined partitions beforehand. This involved changing the logic of how noise was distributed across partitions in the library and ensuring that updated function changes would not break existing function calls. Through this added functionality, developers could also achieve $(\epsilon, 0)$-DP, a stronger privacy guarantee.

While developing features for anonymization libraries, I also became curious about the laws and policies that drove this work. The EU GDPR's carve-outs for anonymized data were enacted in 2020, the very same year that I was hired onto Google's anonymization team. Moreover, anonymization protections, such as differential privacy, are not typically run to increase privacy protections for already-collected data. Rather, they are viewed as techniques to access new forms of data. Then, the very laws that created incentives for the widespread corporate development of anonymization libraries also provide opportunities for anonymized data to violate, rather than preserve, privacy. This potential misalignment between technical and policy values frustrated me, and I began to consider how intended legal values could be more meaningfully incorporated into technical systems.

Models for Legal and Policy Frameworks. I related legal values to technical systems through an undergraduate research project modeling ethical dilemmas in strategic games. In this project, I framed and motivated the system's modeling of autonomous vehicle dilemmas, designed the sacrifice function, and demonstrated how the logical system could be tuned to model different types of ethical and policy frameworks. These contributions led to a co-authored AAAI 2021 publication, *Ethical Dilemmas in Strategic Games*. Yet, models can only be built to capture meaningful aspects of a phenomenon, not every aspect. Our model forced a dilemmic framing between actions, where there might have been a way to incorporate all of them. We also simplistically incorporated legal and ethical frameworks through assigning values for each action and ordering them. Simplification was necessary to make the system computable, and heuristics like ordering can be surprisingly effective. However, law heavily prizes contextual considerations and our technical approach simplified aspects of the problem where contex-

[1]Consider a histogram displaying customers at a restaurant during its publicly available hours. If there are very few customers at 7:00AM, the histogram partition "7:00AM" would likely be dropped with δ-thresholding. This is undesirable because a restaurant's opening hours is publicly available information that can be retained while preserving privacy.

tual legal insight could have been crucial. I was driven to figure out the aspects of AI and technology more broadly that were relevant and meaningful to law.

Uncovering Legal Tensions in the Development of Computational Systems. To better understand relevant technical properties in legal assessments, I conducted research on technology case law with Senior Research Scientist Alice Xiang at Sony AI. I noticed that almost all United States litigation involving facial processing technologies (FPT) was brought through a very specific law in the state of Illinois: the 2008 Biometric Information Privacy Act (BIPA). This led me to investigate the legal properties of BIPA and the technical properties of FPT that combined to create an explosion of litigation. In certain cases, the application of BIPA to FPT highlighted the limitations of technical methods in achieving legal privacy values. However, the policy did not always get to the heart of technical approaches either. For example, agnostic of use case, BIPA protects the collection of scans of face geometry, and excludes facial photographs from privacy protections. This could misguidedly disincentivize privacy-preserving face blurring in image datasets where facial landmarks are collected to detect and blur faces. This is one of several tensions I discuss in my resulting first-author paper *Regulating Facial Processing Technologies: Tensions Between Legal and Technical Considerations in the Application of Illinois BIPA* in the proceedings of ACM FAccT 2022. After the paper was published, the American Civil Liberties Union successfully wielded the policy to bring suit against Clearview AI, and Meta was hit with BIPA lawsuits regarding its augmented reality filters. These lawsuits touch on several points that I raise in the paper, and I was heartened to see BIPA regulate surveillant and discriminatory FPT deployment. At the same time, after uncovering the policy's technical misunderstandings through this research project, I became motivated to consider how written regulation could be improved through a greater incorporation of technological guidance.

Motivating and Proposing Technology Regulation and System Standards. I began conducting research on regulatory design for AI systems in the first year of my master's program advised by Professor Dylan Hadfield-Menell at MIT CSAIL. The AI research community recognizes the importance of transparency regarding potential harms caused by AI systems. However, creating legal and organizational incentives to prioritize investigating these harms remains an open and urgent problem. I discovered the potential of regulatory penalty defaults to address this problem as I synthesized several bodies of literature: AI-related policies in the United States, the legal literature on regulatory design, and domain-specific properties of recommender systems and autonomous vehicles. My main contributions were in the proposal of regulatory penalty default approaches for AI systems. I am the lead author on the corresponding AAAI/ACM AIES 2022 publication *A Penalty Default Approach to Preemptive Harm Disclosure and Mitigation for AI Systems*. The domain-specific, top-down approach I took to this policy problem led me to complementarily explore bottom-up approaches such as developing technical standards. I am currently conducting this work through a part-time research internship at the Center for Trustworthy and Responsible AI within the National Institute of Standards and Technology (NIST), where I am developing and comparing measurable standards for the AI Risk Management Framework.

Future Work: Connecting Technical and Legal Approaches. I am excited that Brown University's research groups interface with the societal impact of technical work. In connecting approaches between computation and law, I would be eager to collaborate with faculty within the Computing for the People project. Within this group, I would be excited to leverage my research background in technology policy and develop research skills in technical approaches for assessing and modelling discrimination, bias, and other policy processes. I am particularly inspired by the work of Professors Suresh Venkatasubramanian and Seny Kamara, as well as PhD student Lizzie Kumar. I am inspired by Professor Venkatasubramanian's work and mentorship in the FAccT research community, as well as his authorship on the AI Bill of Rights–one of the documents I am relating the NIST AI Risk Management Framework to. I am also excited by Professor Kamara's work in privacy and cryptography that is driven from applications and the public interest. I hope to use my doctoral education toward becoming a faculty member conducting research and teaching at the intersection of computer science and law.